

A Universal Assay for Making DNA, RNA, and RNA–DNA Hybrid Configurations for Single-Molecule Manipulation in Two or Three Steps without Ligation

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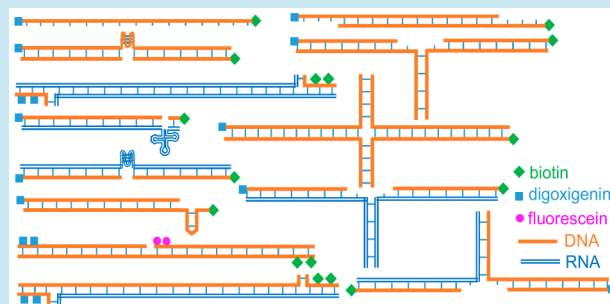
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Supporting Information

ABSTRACT: Despite having a great variety of topologies, most DNA, RNA, and RNA–DNA hybrid (RDH) configurations for single-molecule manipulation are composed of several single-stranded (ss) DNA and ssRNA strands, with functional labels at the two ends for surface tethering. On this basis, we developed a simple, robust, and universal amplification-annealing (AA) assay for making all these configurations in two or three steps without inefficient digestion and ligation reactions. As examples, we made ssDNA, short ssDNA with double-stranded (ds) DNA handles, dsDNA with ssDNA handles, replication-fork shaped DNA/RDH/RNA, DNA holiday junction, three-site multiple-labeled and nicked DNA, torsion-constrained RNA/RDH, and short ssRNA with RDH handles. In addition to single-molecule manipulation techniques including optical tweezers, magnetic tweezers, and atomic force microscopy, these configurations can be applied in other surface-tethering techniques as well.

KEYWORDS: single-molecule, optical tweezers, magnetic tweezers, tethered particle motion tracking, single-stranded DNA, DNA, RNA, molecular motor



To perform diverse functions, nucleic acids (NA), including DNA, RNA, and RNA–DNA hybrid (RDH), exist in diverse topologies in cells. In order to understand the properties and functions at the single-molecule level, various NA configurations have been excessively studied using single-molecule manipulation techniques such as optical tweezers, magnetic tweezers (MT), and atom force microscopy (AFM).^{1,2} The proteins and other factors have been studied using single-molecule manipulation techniques as well. The manipulated DNA structures include single-stranded DNA (ssDNA),^{3–16} G-quadruplex,^{17–24} i-motif,²⁵ origami,^{26,27} short hairpin,^{28–35} double-stranded (ds) DNA with ssDNA handles,³⁶ replication-fork-shaped (Y-shaped) DNA,^{37–43} DNA holiday junction (HJ DNA),^{44,45} three-site multiple-labeled and nicked DNA,^{46–50} end-opened DNA,^{51–56} end-closed and torsional-unconstrained DNA,^{55–58} and torsional-constrained DNA.⁵⁹ In these works, repetitive digestion and ligation reactions were usually needed every time the DNA configurations were made, which was very time-consuming and inefficient. After digestion or ligation, gel purification was often required, during which DNA nicks might occur. In order to avoid using the inefficient ligation reactions, an approach

based on megaprimer PCR was developed to make the torsion-constrained DNA.⁵⁹ Also in order to avoid ligation, an approach based on site-specific nicking was developed to make ss-ds DNA junctions.^{60–62} These developed approaches, however, are limited to the preparation of certain DNA configurations.

The manipulated RNA configurations include short hairpin,^{63–70} pseudoknot,^{71,72} kissing loop,⁷³ G-quadruplex,^{74,75} riboswitch,^{76–78} and ribozyme.⁷⁹ In these works, the short ssRNA with long RDH handles were made through annealing the transcription-amplified ssRNA to PCR-amplified dsDNA fragments. Because the RDH duplex formed only if the homologous ssDNA strand in the dsDNA duplex was completely replaced by the ssRNA, a tough and time-consuming annealing procedure using formamide was needed.⁸⁰ In addition, the mechanical and structural properties of RNA duplex were studied by single-molecule manipulation also.^{53,81–83}

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Despite having a great variety of topologies, most NA configurations with labeled ends for single-molecule manipulation are composed of long ssDNA and ssRNA. On this basis, we aim to develop an amplification-annealing (AA) assay for making all these NA configurations in two or three steps without the inefficient digestion and ligation reactions. In the AA assay, we amplify the long ssDNA and ssRNA through one-sided PCR (OSP) and *in vitro* transcription, respectively, and directly anneal them together.

RESULTS

ssDNA Configuration. The ssDNA configuration was excessively applied to study the mechanical properties of ssDNA as well as its interaction with proteins using single-molecule manipulation techniques.^{3–16} Here, we made it through OSP directly (Figure 1a). We labeled its 5' end with a

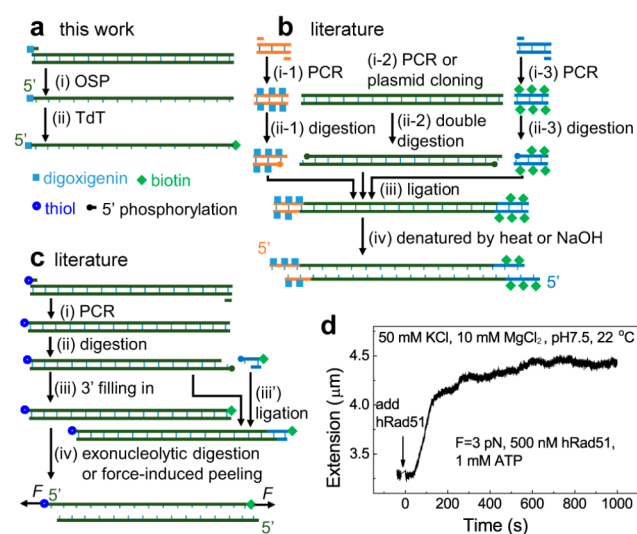


Figure 1. ssDNA configuration. (a–c) Flow-charts to make it in this work (a) and literature (b,c). (d) Dynamics of human Rad51 polymerization on ssDNA at a constant force. During the formation of Rad51-ssDNA nucleoprotein filament (~ 0.51 nm per nucleotide), the extension of the stretched ssDNA elongated gradually under the constant force of 3 pN.

digoxigenin group using the functionalized OSP primer (step i) and labeled its 3' end with a biotin group using terminal deoxynucleotidyl transferase (TdT) and biotin-dUTP (step ii). Stretched by MT, the dynamics of human Rad51 polymerization on a single ssDNA was consistent with previous studies^{5,11} (Figure 1d).

Typically, two methods were used to make the ssDNA for single-molecule manipulation. (1) Ligation of two multiple-labeled short DNA fragments to the two ends of a long DNA fragment, utilizing their endonuclease-generated sticky ends. Then, the dsDNA was denatured by heat or/and NaOH^{3–9} (Figure 1b). (2) Labeling both the 5' and 3' ends of the same strand in the dsDNA at the terminals, and then getting rid of the unlabeled DNA strand by exonucleolytic digestion^{10–13} or force-induced peeling after surface tethering^{14–16} (Figure 1c). The 5' end was labeled using a functionalized PCR primer, and the 3' end was labeled either by 3' filling in with functionalized dNTPs or ligation to the 3' functionalized synthesized oligos. In contrast with the two existing methods that contain four reaction steps, our AA assay needs only two steps and avoids the inefficient digestion and ligation reactions.

Short ssDNA with dsDNA Handles. The short ssDNA can form important functional structures, such as G-quadruplex,^{17–24} i-motif,²⁵ origami,^{26,27} and short hairpin.^{28–35} Here, we made the short ssDNA with two or one dsDNA handle (ds-ss-ds-DNA or ds-ss-DNA) through annealing together three or two OSP-amplified ssDNA strands (Figure 2a,b). We labeled its two ends using the functionalized OSP primers (step i).

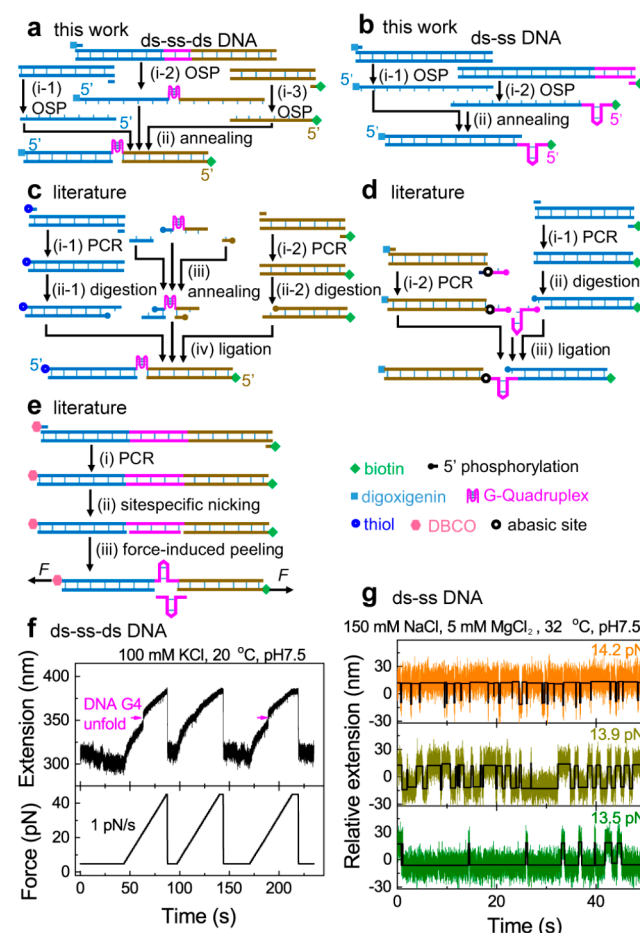


Figure 2. Short ssDNA with dsDNA handles. (a,b) Flow-charts to make the ds-ss-ds-DNA (a) and ds-ss-DNA (b) in this work. (c–e) Flow-charts to make the ds-ss-ds-DNA drawn according to the descriptions in refs 18 (c), 31 (d), and 62 (e). (f) The force-induced unfolding of the DNA G-quadruplex at a loading rate of 1 pN/s. (g) Hopping dynamics of the 26-bp DNA hairpin at constant forces and constant buffer conditions measured using the same DNA molecule. The time-courses are analyzed using the Hidden Markov model (black steps).

As shown in Figure 2a, we chose the DNA molecule made through the traditional PCR-digestion-ligation procedure as the OSP template to generate the ssDNA containing the G-quadruplex forming sequences.¹⁸ Once the dsDNA template has been prepared, we simply amplified it by the AA assay rather than repeating the PCR-digestion-ligation steps afterward. Stretched by MT, the unfolding dynamics of our G-quadruplex was consistent with the literature¹⁸ (Figure 2f and Figure S1a).

Rather than sandwiched between two dsDNA handles (ds-ss-ds-DNA), the short ssDNA could be placed at one end of the DNA also (ds-ss-DNA). As an example, we made a 26-bp

DNA hairpin with a dsDNA handle through annealing two OSP-amplified ssDNA strands (Figure 2b). Stretched by MT, the DNA hairpin showed a two-state hopping dynamics near the equilibrium unfolding force (Figure 2g), which was consistent with previous results obtained at constant forces.^{33–35}

The typical method to make the ds-ss-ds-DNA was inserting the synthesized short ssDNA into two dsDNA handles utilizing their endonuclease-generated sticky ends^{17–30} (Figure 2c). The efficiency of three-way ligation can be improved by generating a long 5' sticky end of the dsDNA handle through autosticky PCR^{31–35} (Figure 2d). To avoid the inefficient three-way ligation, a double site-specific nicking assay was developed, where the short complementary ssDNA between two nicks was peeled off by force after surface-tethering⁶² (Figure 2e). Compared to these three methods, our two-step AA assay is still the simplest.

dsDNA with ssDNA Handles. Opposite to the ds-ss-ds-DNA, a dsDNA fragment sandwiched between two ssDNA handles (ss-ds-ss-DNA) was applied to prevent the interactions between the inside dsDNA and the substrate surfaces in the presence of multivalent cations.³⁶ In that work, the ss-ds-ss-DNA was made through a complex procedure including reaction steps of site-specific nicking, dephosphorylation, endonucleolytic digestion, 3' filling-in, and 5' → 3' enucleation (Figure 3b). Here, we made the ss-ds-ss-DNA in two steps through annealing together two OSP-amplified ssDNA strands (Figure 3a). Stretched by MT, the force–extension curves of

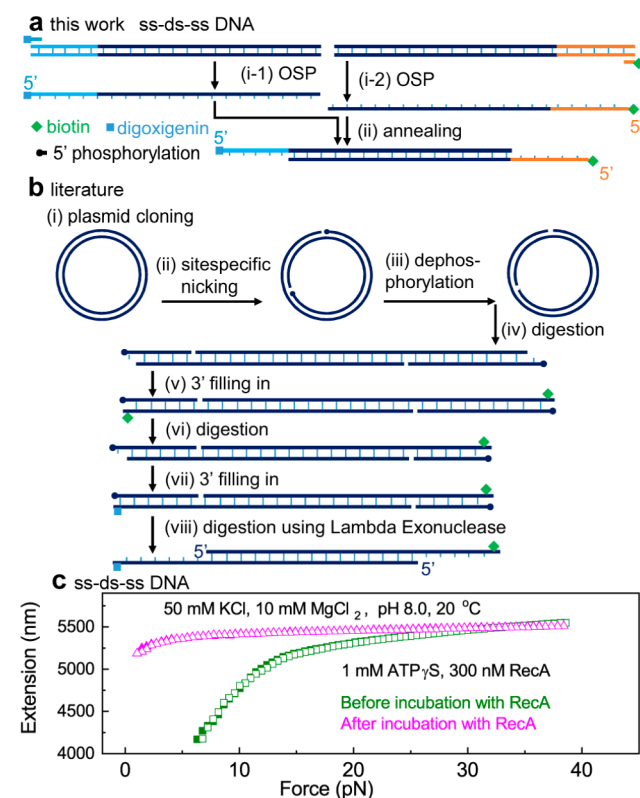


Figure 3. dsDNA with ssDNA handles. (a) The flow-chart to make the ss-ds-ss DNA in this work. (b) The flow-chart to make the ss-ds-ss-DNA drawn according to the description in ref 36. (c) The elasticity of the ss-ds-ss-DNA before (olive squares) and after (magenta triangles) formation of RecA-ssDNA nucleoprotein filaments measured using the same DNA molecule.

the ss-ds-ss-DNA before and after the formation of RecA-ssDNA nucleoprotein filament were shown for comparison (Figure 3c), which was consistent with the elasticity of RecA-ssDNA filaments obtained previously.^{6,14}

Replication-Fork-Shaped DNA. The Y-shaped DNA was often applied to study the force-induced unzipping of DNA duplex as well as the functions of DNA helicases for single-molecule manipulation.^{37–43} Here, we made the Y-DNA in two steps through annealing together four OSP-amplified ssDNA strands (Figure 4a): a 5' digoxigenin-labeled ssDNA (strand 1), a 5' biotin-labeled ssDNA (strand 3), and two unlabeled ssDNA (strands 2 and 4). It contained 5' and 3' loading sites for DNA helicases. Stretched by MT, the force–

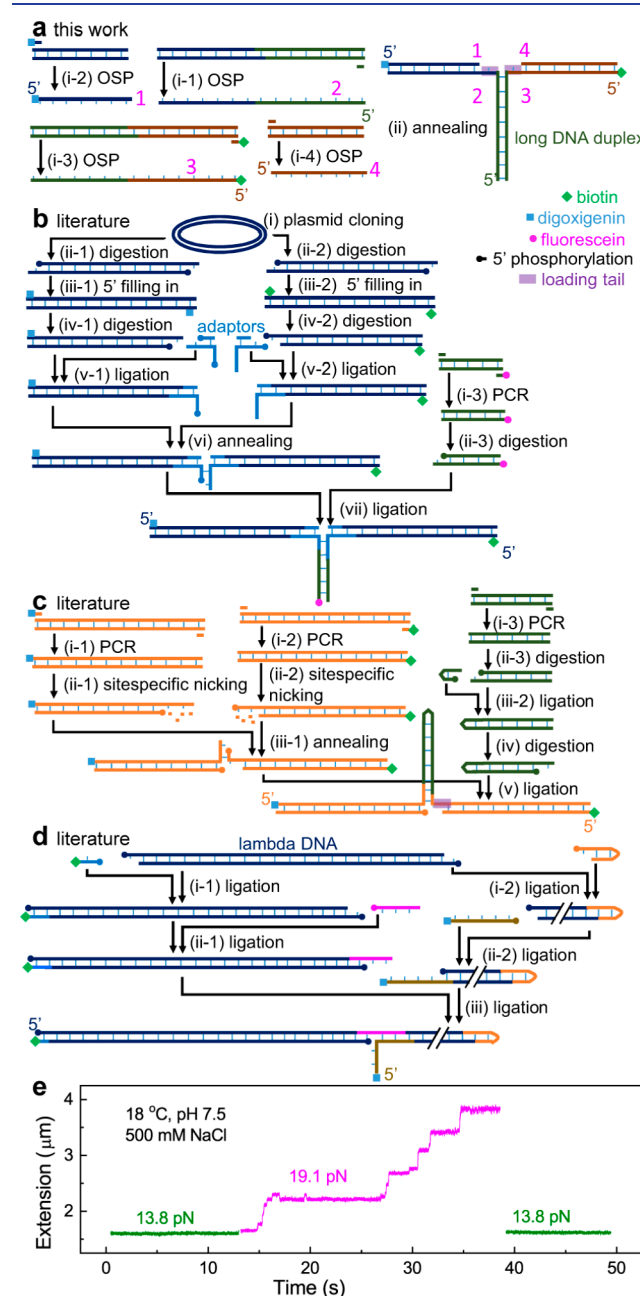


Figure 4. Y-shaped DNA. (a) The flow-chart to make it in this work. (b–d) Flow-charts to make it drawn according to the descriptions in refs 37 (b), 38 (c), and 40 (d). (e) Unzipping dynamics of the Y-shaped DNA.

induced unzipping of the DNA duplexes showed stepwise kinetics at the constant force (Figure 4e), which may be due to the sequence-dependent energy barriers along with DNA unzipping.^{40–42}

The typical method to make the Y-DNA was ligation of three long DNA fragments mediated by synthesized adaptors for the endonuclease-generated sticky ends of long DNA fragments³⁷ (Figure 4b). An alternative way to the sticky end adaptors was using site-specific nicking to generate long overhangs at the unlabeled ends of dsDNA handles^{38,39} (steps ii-1 and ii-2, Figure 4c). In order to minimize the inefficient digestion and ligation reactions, the procedure could be greatly simplified utilizing the 12-nucleotides sticky ends of lambda DNA, but the DNA sequences were restricted^{40–43} (Figure 4d). In comparison with these three existing methods, our two-step AA assay is the simplest without the inefficient ligation reactions.

DNA Holiday Junction. The HJ DNA is a key intermediate in homologous recombination, whose functions were studied by single-molecule manipulation.^{44,45} Here, we made the HJ DNA in two steps through annealing together four OSP-amplified ssDNA strands (Figure 5a): a 5' digoxigenin-labeled ssDNA (strand 3), a 5' biotin-labeled ssDNA (strand 2), and two unlabeled ssDNA (strands 1 and 4). The two long wing-like duplexes (cyan and orange) can be unzipped synchronously by external stretching force, which causes an increase in the tether extension. As shown in Figure 5d, using MT, the HJ DNA was unzipped at 36.5 pN that was about 2-fold of the unzipping force of a single DNA hairpin. Besides for studying the biological functions, the HJ unzipping is also a revisable molecular force-sensor which responds to forces greater than DNA hairpin unzipping force (12–20 pN) and less than DNA overstretching force (50–70 pN).

Typically, the HJ DNA was made by inserting a short HJ core into two dsDNA handles utilizing their endonuclease-generated sticky ends (Figure 5b,c). The synthesized oligos were either ligated to the handles one by one⁴⁴ (Figure 5b) or inserted between the two dsDNA handles as a preformed core⁴⁵ (Figure 5c). Rather than carry out the complex digestion and ligation reactions in the two existing methods, our AA assay requires only two steps.

Three-Site Multiple-Labeled and Nicked DNA. The three-site multiple-labeled and nicked DNA configuration was often applied for rotor bead tracking to study the torsion-related properties of DNA duplex in MT experiments.^{46–50} Here, we made it by three reaction steps (Figure 6a). (i) Amplification of three multiple-labeled short ssDNA (a few hundreds of nucleotides long) through OSP. (ii) Amplification of three long ssDNA through OSP using the multiple-labeled ssDNA as megaprimers. Thus, the 5' end of these long ssDNA were multiple-labeled. (iii) Annealing together the three long ssDNA strands. As shown in Figure 6d, the DNA was stretched using our transverse MT. Unfortunately, our MT is not capable of rotor bead tracking.

Typically, this DNA configuration was made by a digestion-ligation procedure^{46–50} (Figure 6b). To leave a nick, a dephosphorylation reaction was required before ligation (step iii). Overall, the DNA configuration was prepared by ligating five digested DNA fragments, which was very inefficient. In order to improve the efficiency, an approach based on site-specific nicking was developed, where the internal nucleotides between nicking sites were replaced by a synthesized multiple-labeled oligo^{60,61} (Figure 6c). In comparison with these two

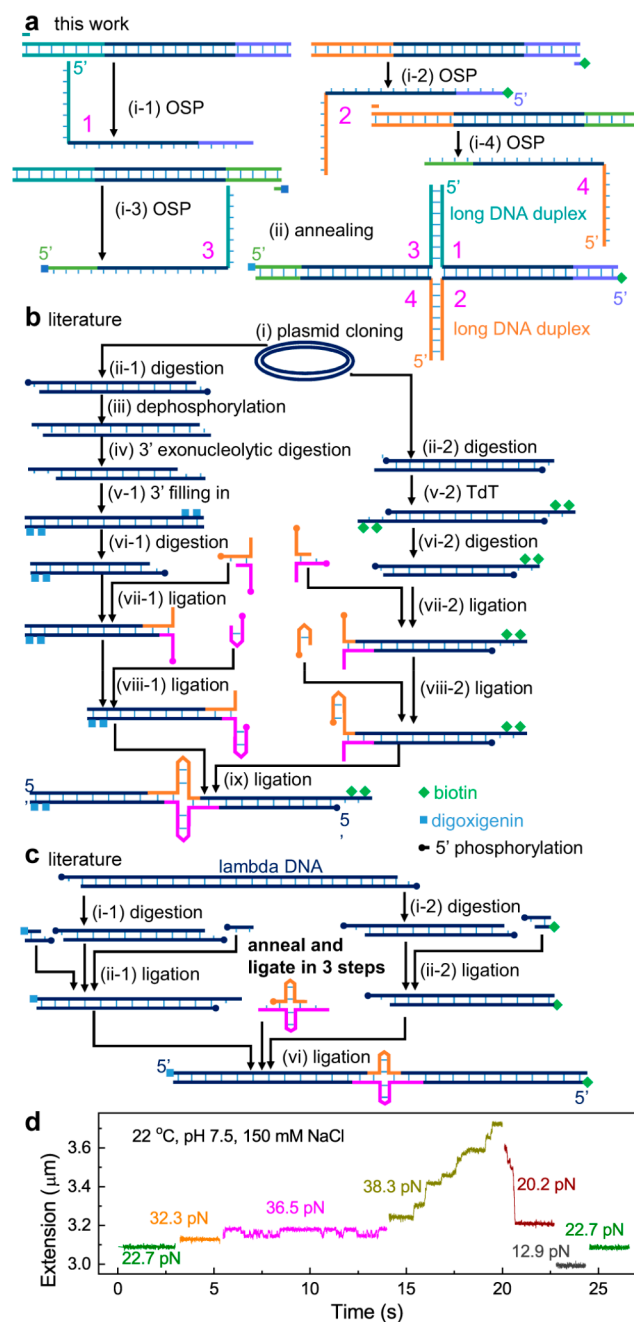


Figure 5. DNA holiday junction. (a) The flow-chart to make it in this work. (b,c) Flow-charts to make it drawn according to the descriptions in refs 44 (b) and 45 (c). (d) Dynamics of HJ unzipping and rezipping at constant forces.

existing methods, our three step AA assay is still the simplest. If synthesized multiple-labeled oligos were used as megaprimers, the AA assay needs only two steps (steps ii–iii) to make this DNA configuration.

RNA Structures with RDH Handles. In the literature, the folded RNA structures with RDH handles were made through annealing the ssRNA to dsDNA at tough conditions (higher than 95 °C) with formamide.^{63–79} Here, we successfully annealed ssRNA with ssDNA at a gentle condition (65 °C) without formamide. As an example, we made a G-quadruplex forming short ssRNA sandwiched between two long RDH handles through annealing the transcription-amplified ssRNA with two OSP-amplified ssDNA (Figure 7a). We used MT to

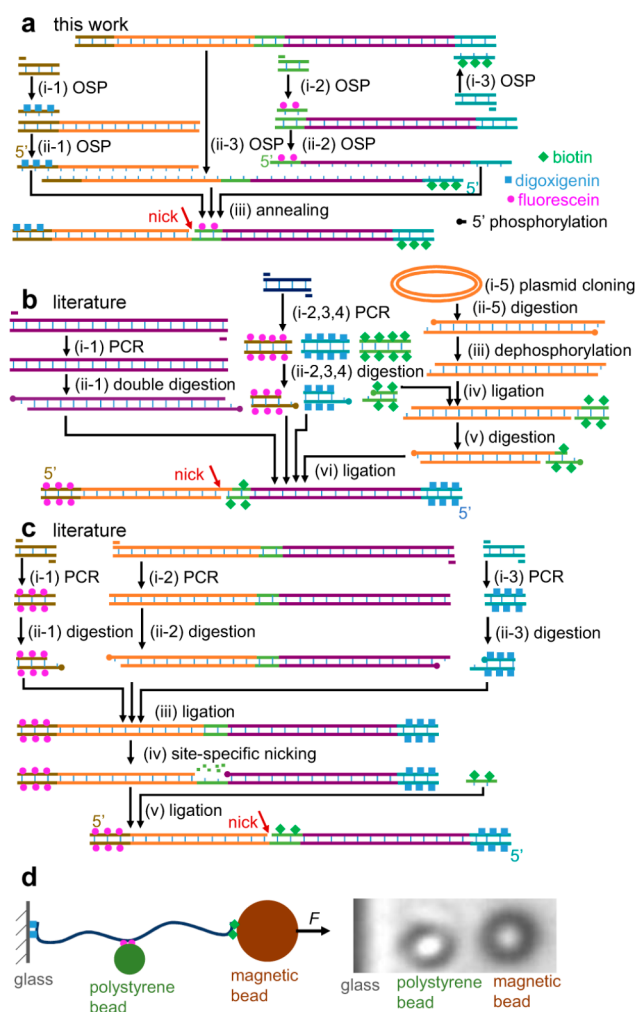


Figure 6. Three-site multiple-labeled and nicked DNA. (a) The flow-chart to make it in this work. (b,c) Flow-charts to make it drawn according to the descriptions in refs 46 (b) and 60 (c). (d) In our transverse MT, the DNA was tethered between a glass surface and a superparamagnetic microbead. An anti-fluorescein coated polystyrene bead was attached to the internal fluorescein labeled site of the anchored DNA molecule.

unfold the RNA G-quadruplex at 1 pN/s, where similar results were obtained as in the literature⁷⁴ (Figure 7c and Figure S1b). As another example, we made a guanine riboswitch sandwiched between a long RDH handle and a short RDH handle through annealing the ssRNA with a 5' digoxigenin-labeled OSP-amplified ssDNA and a 3' biotin-labeled synthesized DNA oligo (Figure 7b). Stretched by MT, the riboswitch was unfolded and refolded during the stretch-relax force cycle in the presence of 1 μ M guanine, which was consistent with the previous results⁷⁶ (Figure 7d).

Torsion-Constrained RNA. The mechanical properties of dsRNA were studied at the single-molecule level recently.^{53,81–83} The torsion-constrained DNA can be made directly by PCR using multiple-labeled short DNA fragments as megaprimers.⁵⁹ Its closest cousin, the torsion-constrained RNA, however, is hard to be made. Here, we made it through annealing together two long transcription-amplified ssRNA strands and two short OSP-amplified multiple-labeled ssDNA strands, where the breaks between DNA and RNA strands were merged together by ~ 30 RDH base pairs (Figure 8a). Twisted by MT, the torsion-constrained RNA demonstrates

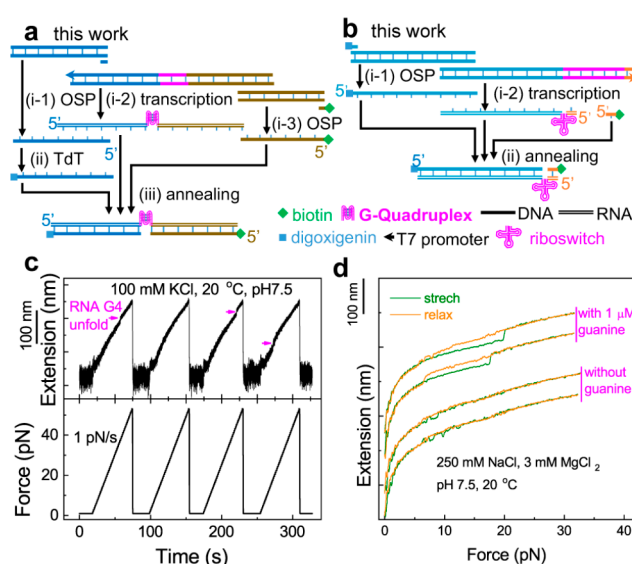


Figure 7. RNA structures with RDH handles. (a) The flow-chart to make the RNA structure with two long RDH handles in this work. (b) The flow-chart to make the RNA structure with a long RDH handle and a short RDH handle in this work. (c) The force-induced unfolding of RNA G4 at a loading rate of 1 pN/s. (d) The force-induced unfolding of the guanine riboswitch at a loading rate of 1 pN/s with or without guanine in the solution measured using the same molecule.

typical bell-shaped twist-extension curves due to the formation of plectonemes when the duplexes are positively or negatively supercoiled (Figures 8d), which was consistent with the previous results.⁸³

Previously, there were two methods to make the torsion-constrained RNA. (1) Ligation of a long dsRNA fragment to two multiple-labeled short dsRNA handles. All of the three dsRNA fragments were generated by annealing two transcription-amplified ssRNA followed by the 5' monophosphatation reactions^{81,82} (Figure 8b). In their experiments, only less than 1% of the ligated molecules were found to be torsion-constrained.⁸² (2) Annealing two 5' multiple-labeled ssRNA strands generated by two-step transcriptions, where special materials were required to purify the partial transcription complex⁸³ (Figure 8c). Distinct from these two existing methods, our two-step AA assay is much simpler and needs no special tool enzymes or special materials. We found more than 20% of the AA-generated molecules were torsion-constrained.

A Few of the New NA Configurations. In addition to the existing NA configurations, we made a few new ones using the AA assay, which should be helpful for single-molecule manipulation. As shown in Figure 9a, we made the Y-shaped RDH and RNA which are ideal to study the force-induced unzipping or helicases unwinding of RDH or RNA duplex. We made the Y-shaped RDH and RNA through annealing together four ssDNA/ssRNA strands (Figure 9a): a 5' digoxigenin-labeled ssDNA (strand 1), a 3' biotin-labeled ssDNA (strand 4), an unlabeled ssDNA or ssRNA (strand 2 or 2'), and the other unlabeled ssRNA (strand 3'). The Y-shaped RDH and RNA contain 5' and 3' single-stranded loading sites for helicases and thousands of base pairs to be unwound. Stretched by MT, the unzipping/rezipping dynamics of the RDH and RNA duplexes showed that the unzipping force of RNA was significantly larger than that of RDH at the same

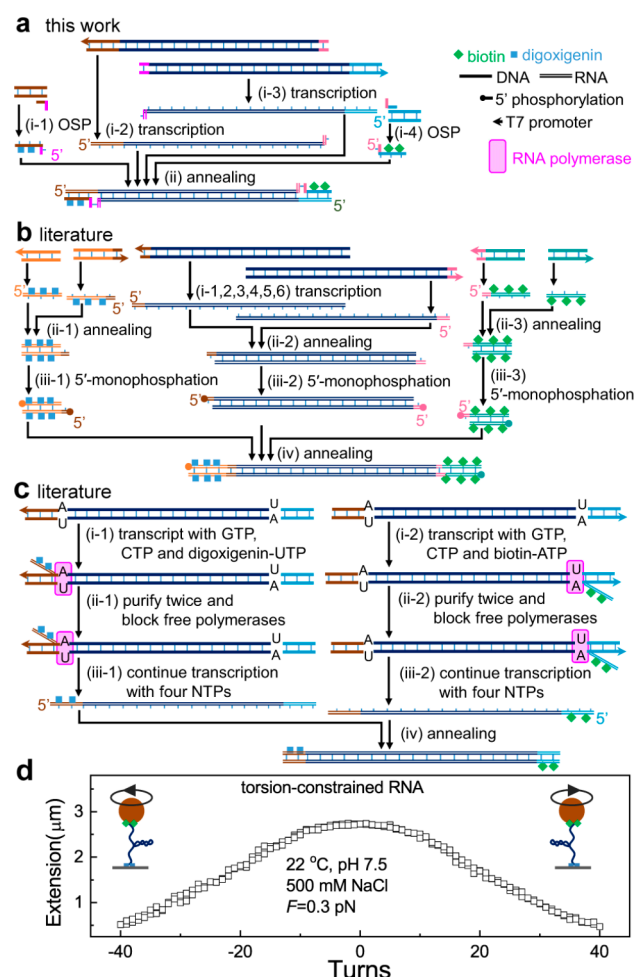


Figure 8. Torsion-constrained RNA. (a) The flow-chart to make it in this work. (b,c) Flow-charts to make it drawn according to the descriptions in refs 82 (b) and 83 (c). (d) The twist-extension curve of the torsion-constrained RNA measured using MT.

buffer conditions (Figure 9c,d), which may be due to the higher thermal stability of RNA–RNA base pairs.⁶⁹

The other new NA configuration was the torsion-constrained RDH, which is ideal to study the torsional-related properties of RDH duplex (Figure 9b). Similar to the torsion-constrained RNA (Figure 8a), we made it by annealing together a long ssRNA, a long ssDNA, and two short multiple-labeled ssDNA. Twisted by MT, a bell-shape twist-extension curve was obtained probably due to the formation of plectonemes⁸³ (Figure 9e).

DISCUSSION

In this work, we have developed an AA assay to make various DNA, RNA, and RDH configurations for single-molecule manipulation. By breaking down the operating units from dsDNA fragments to ssDNA strands, we got much more freedom to make the NA configurations in a better way. For surface tethering, the NA configurations should be labeled at the two ends. In the AA assay, we label the 5' end of an OSP-amplified ssDNA using the functionalized primer (Figure 1a, 2a,b, 3a, 4a, 5a, 7a,b, and 9a,b). If 3' labeling is required, we label it using TdT with functionalized dNTPs (Figure 1a, 7a, and 9a). If multiple labeling is required, we amplify a multiple-labeled short ssDNA with functionalized dNTPs and use the

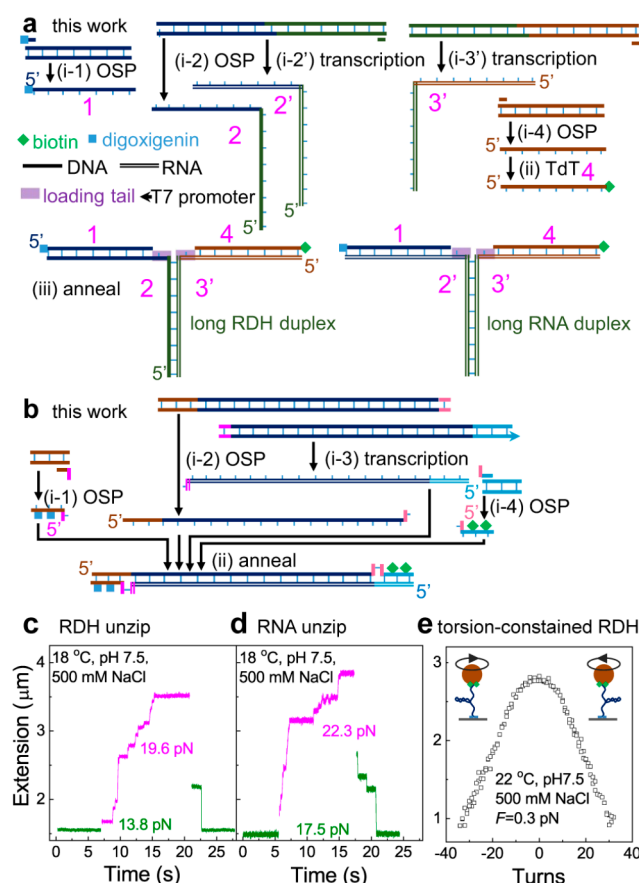


Figure 9. A few of the new NA configurations. (a) The flow-chart to make the Y-shaped RDH and RNA. (b) The flow-chart to make the torsion-constrained RDH. (c,d) The force-induced unzipping of the long RDH (c) and RNA (d) duplexes. (e) A twist-extension curve of the torsion-constrained RDH measured using MT.

short ssDNA as a megaprimer (Figure 6a). Rather than make a covalently continuous multiple-labeled ssRNA, we use multiple-labeled short ssDNA as handles to make the torsion-constrained RNA/RDH (Figure 8a and 9b).

The AA assay has quite a few advantages.

- (1) Universality. Although the NA configurations for single-molecule manipulation have diverse topologies, generally they are composed by ssDNA and ssRNA strands. Thus, we just amplify these ssDNA and ssRNA strands and anneal them together. The AA assay can be used to make various NA configurations, such as ssDNA, ds-ss-DNA, ds-ss-DNA, ss-ds-ss-DNA, Y-shaped DNA/RDH/RNA, HJ DNA, three-site multiple-labeled and nicked DNA, torsion-constrained RNA/RDH, as well as ssRNA with RDH handles.
- (2) Time saving. The existing methods often require four to nine reactions and require gel purifications between the reactions. Thus, the total time cost is about several days. The AA assay only requires two or three steps, which can be completed in several hours.
- (3) High yield. Because the inefficient ligation reactions are avoided, the gel purifications are minimized, and fewer reaction steps are required, the AA assay has a relatively high yield.
- (4) Fewer tool enzymes. The only tool enzymes used in the AA assay are the DNA/RNA polymerases and terminal

deoxynucleotidyl transferase. Those special tool enzymes and special materials are not needed.

- (5) Easy to design. The existing assays generally need to design the recognizing sites for tool enzymes, the complementary sticky ends for ligation, and a complex flowchart. These messy designs are avoided in the AA assay.
- (6) High fidelity. Different from the ordinary two-sided PCR where the misincorporated mutations may accumulate each thermal cycle, in the OSP all the nucleotides are incorporated according to the original DNA template. Thus, the OSP intrinsically has higher fidelity than the ordinary two-sided PCR. In this work, we used KOD FX DNA polymerase for OSP, which might misincorporate a wrong nucleotide about every five thousand nucleotides. Since the short ssDNA strands we studied were only tens of nucleotides long, the misincorporation has no effect on our results. For those studies needing even higher fidelity, ultrahigh fidelity DNA polymerases should be used.

For those linear DNA configurations, such as end-opened DNA,^{51–56} end-closed and torsional-unconstrained DNA,^{55–58} and torsional-constrained DNA,⁵⁹ they are already easy to be made in two or three steps. Thus, although capable, we did not try to make them using the AA assay. In our experiments, we found occasionally some AA-generated molecules had wrong extensions which were discarded. The molecules with the correct extensions mostly gave the expected signals, which confirmed they had correct configurations.

In this work, we generated the thousands of nucleotides long ssDNA molecules by OSP and generated the thousands of nucleotides long ssRNA by *in vitro* transcription. Although quite high in price, these long ssDNA and ssRNA molecules can be ordered from several companies or directly made using a synthesizer. If the ssDNA and ssRNA are chemically synthesized directly, all NA configurations in this work can be made in only one annealing step.

In this work, we chose MT as the primary tool to demonstrate the functionality of the AA-generated NA configurations. However, it should be possible to apply them in other surface-tethering techniques as well, such as optical tweezers, AFM, flow stretching, DNA microarray, and tethered particle motion tracking. If manipulation and fluorescence microscopy are combined, it should also be possible to stretch the NA configurations and track directed motion of molecular motors or diffusion of other proteins on it.

METHODS

One-sided PCR, also called single primer PCR, is a variation of PCR to amplify one strand of the original DNA with the primer at only one side. The product of OSP is ssDNA and its productivity is generally lower than the ordinary two-sided PCR. One reason that OSP is not popularly used is the difficulty to amplify long ssDNA. We overcame this difficulty by using more dsDNA templates (20 ng/ μ L) and less thermal cycles (15–20 cycles) than the two-sided PCR. The quantitative effects of the concentration of dsDNA template and the number of thermal cycles on OSP are shown in Figure S2. If in need, the OSP-generated long ssDNA can be purified by agarose gel electrophoresis (Figure S3).

After OSP and transcription, we added EDTA into the reaction systems to a concentration of 30 mM without

purification or dilution. Then, we annealed the complementary ssDNA/ssRNA through a 2 h process containing a 1 h incubation step at 65 °C followed by a 1 h cooling process from 65 to 40 °C. During the preparation of each NA configuration, we performed agarose gel electrophoresis to monitor the process step by step, where the final products were confirmed (Figure S4, S5, and S6).

We used Taq DNA polymerase (Thermo Fisher Scientific) for modified dNTPs and KOD FX DNA polymerase (Toyobo) for unmodified dNTPs in PCR. We purchased human Rad51 from Abcam and all the other tool enzymes from New England Biolabs.

We used home-built MT to manipulate the NA configurations.^{53–57} We anchored the digoxigenin-labeled end of the NA molecule to an antidigoxigenin coated cover glass and anchored the biotin-labeled end to a streptavidin-coated superparamagnetic microbead (M-270, Dynabeads). We operated a pair of NdFeB magnets to stretch and twist the anchored molecule in a flow-cell. We put the flow-cell on the top of a Peltier chip to control the inside temperature and connected the flow-cell to a syringe pump to control the inside buffer composition. The experimental temperature was measured using a thermocouple thermometer. We passivated the substrate surfaces using 10% w/v mPEG-Succinimidyl Valerate (MW 2K, Laysan Bio) to avoid nonspecific binding. Please find more details about the home-built MT in our previous publications.^{53–57}

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acssynbio.9b00241.

Figures S1–S6 and Supplementary Methods (PDF)

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X.H.Z. designed this work. Y.J.Y., L.S., X.C.Z., C.Z., H.F., E.C.Z., and W.Q.W. performed the experiments. X.H.Z., Y.J.Y., and H.J.Y. wrote the paper.

Notes

The authors declare no competing financial interest.

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